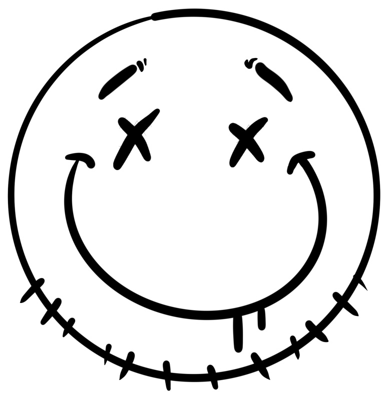

Lab 11: Malware Analysis

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CSEC3755-001

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Contents

1 Executive Summary

2 Triage & File Identification

Note about this section:

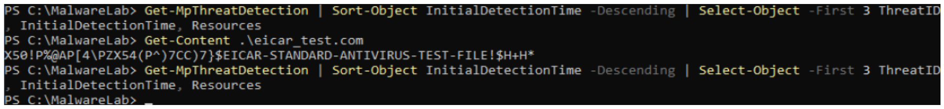
The `Add-WindowsCapability` command for Sysmon that was given did not work for this environment. Instead, I used the following:

```
# Download
Invoke-WebRequest -Uri "https://download.sysinternals.com/files/Sysmon.zip"
    -OutFile "$env:TEMP\Sysmon.zip"

# Extract
Expand-Archive -Path "$env:TEMP\Sysmon.zip" -DestinationPath "C:\Tools\
   Sysmon"

# Install (run this as Admin)
C:\Tools\Sysmon\Sysmon64.exe -accepteula -i
```

2.1 Creating the EICAR Test File (Sample A)

The following screenshots show the process of creating the EICAR test file in the Windows VM. I used the full sting given hoping to see Defender defy my expectations and catch the file at creation, but the file survived create and `Get-Content`

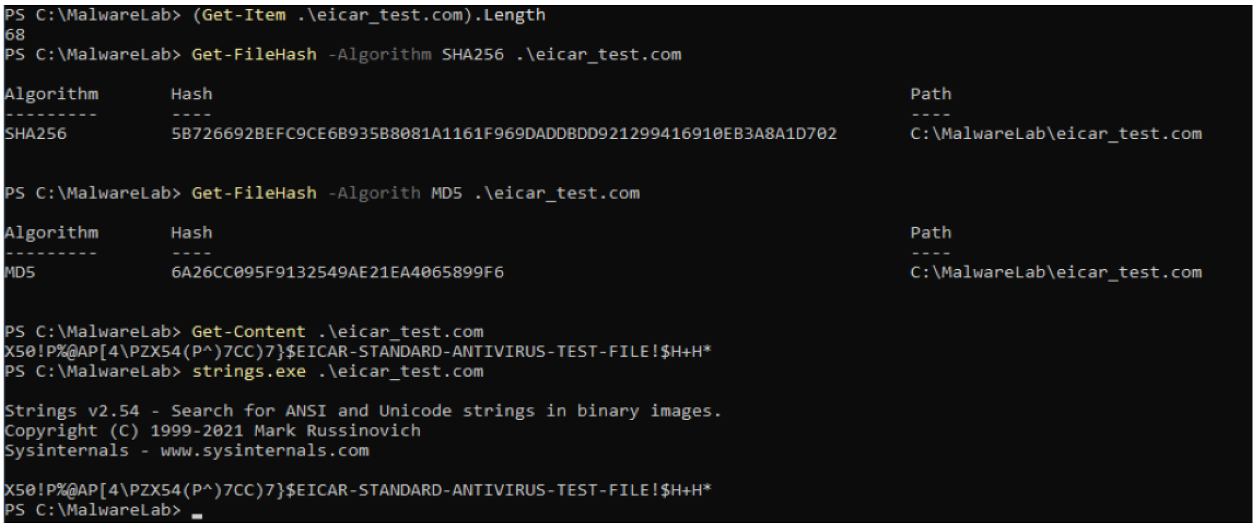
Figure 1.1: Creation of the EICAR test file



```
PS C:\MalwareLab> Get-MpThreatDetection | Sort-Object InitialDetectionTime -Descending | Select-Object -First 3 ThreatID
, InitialDetectionTime, Resources
PS C:\MalwareLab> Get-Content .\eicar_test.com
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
PS C:\MalwareLab> Get-MpThreatDetection | Sort-Object InitialDetectionTime -Descending | Select-Object -First 3 ThreatID
, InitialDetectionTime, Resources
PS C:\MalwareLab> _
```

Figure 1.2: The EICAR test file survives Windows Defender

2.2 Basic File Identification



```
PS C:\MalwareLab> (Get-Item .\eicar_test.com).Length
68
PS C:\MalwareLab> Get-FileHash -Algorithm SHA256 .\eicar_test.com

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          5B726692BEFC9CE6B935B8081A1161F969DADDBDD921299416910EB3A8A1D702       C:\MalwareLab\eicar_test.com


PS C:\MalwareLab> Get-FileHash -Algorith MD5 .\eicar_test.com

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
MD5             6A26CC095F9132549AE21EA4065899F6                                       C:\MalwareLab\eicar_test.com


PS C:\MalwareLab> Get-Content .\eicar_test.com
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
PS C:\MalwareLab> strings.exe .\eicar_test.com

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
PS C:\MalwareLab> _
```

Figure 1.3: Results of file identification commands for the EICAR test file

2.3 Hash Lookup on Virus Total

- Out of 62 security vendors on Virus Total, 6 flagged the file as malicious.
- A few names given for the threat are:
  - `Virus/EICAR_Test_File`
  - `NotAThreat.EICAR[TestFile]`
  - `Eicar_test_1`
- Tags shown for the 'threat' are:
  - powershell
  - idle

  - long-sleeps

- File type according to Virus Total: **Powershell**
- As for the community tab, there isn't any activity early then 10 years.

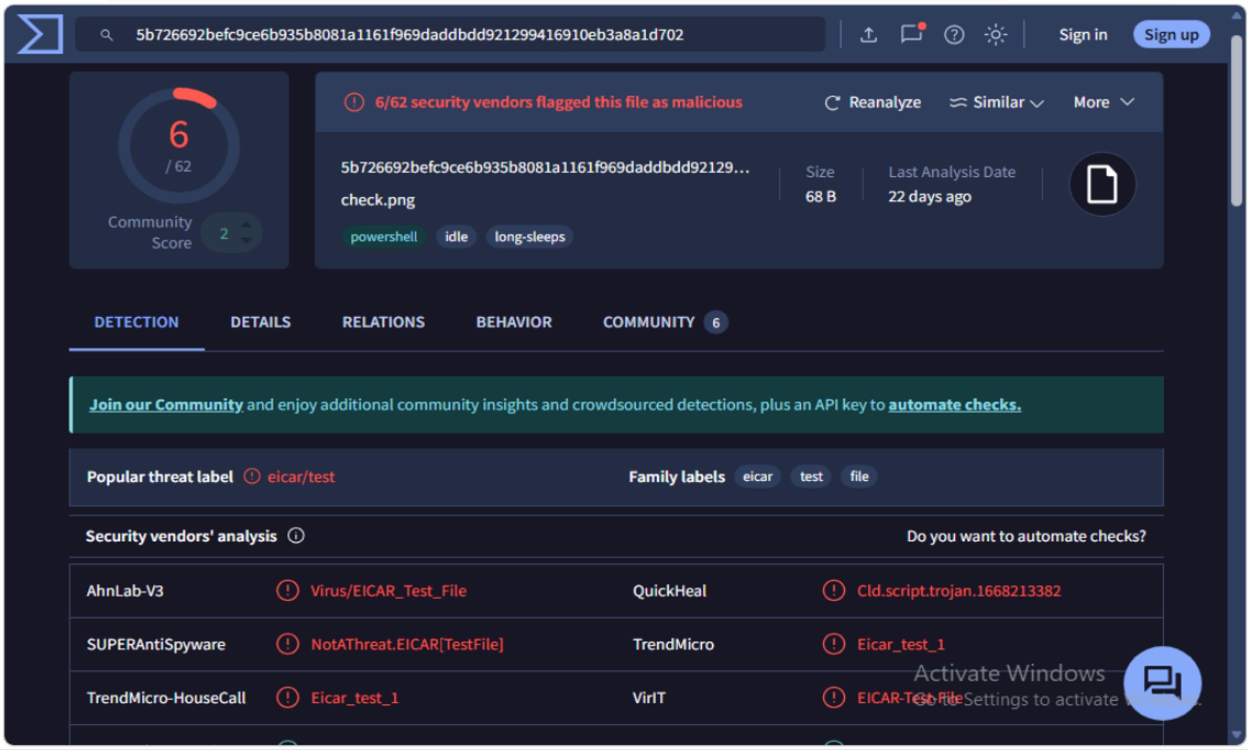


Figure 1.4: Results for Virus Total hash look-up

> It should be noted that while the results on Virus Total showed a 6/62, the only results to show in the 'Security vendors' analysis' table were those from AhnLab-V3, SUPER-AntiSpyware, and TrendMicro-HouseCall.

What kind of threat is this?

This is an idle **Powershell script** that waits until the right conditions are met before executing. This would explain how it was able to evade detection from Windows Defender despite being flagged as malicious.

3 Static Analysis

4 Dynamic Analysis

5 IOC Report & Detection

++report written and formatted with help from Claude (Sonnet 4.6)++